Category Theory for Mathematical Physics — Exercise Sheet 11

Algebras in tensor categories and module categories

Variant: Solutions

Conventions as in Lecture 11: C is a fusion category, $\mathcal{M}$ a left C-module category, $[M,N]$ the internal Hom (Definition 11.7); ψ is a 2-cochain (in Exercise 6, ψ is Ising's fermion instead), and $(\delta\psi)$, ω are as in Lecture 3. The *NIM-rep* of a semisimple module category $\mathcal{M}$ with simple objects $M_1,\dots,M_r$ is the family of matrices $(n_a)_{ij} = \dim \mathrm{Hom}_{\mathcal{M}}(a \otimes M_j, M_i)$, one for each simple a of C. In general, a *NIM-rep* of the fusion ring of C is a family of square non-negative integer matrices n_a, all of one size $r \geq 1$, with $n_1 = 1$ for the unit (n_0 in Exercise 5), $n_a n_b = \sum_c N_{ab}^c\, n_c$ and $n_{a^*} = n_a^T$; it is *indecomposable* if no nonempty proper set of indices (simple objects) is closed under every n_a. Module categories on this sheet are nonzero. $\mathbb{Z}_N$ is additive, $\mathbb{Z}_2 = \{1, s\}$ and S_3 multiplicative. In Exercise 2, ε is Sheet 3's carry (not the toric fermion) and $\zeta_4 = i$. In Exercise 5, U_m is the polynomial with $U_0 = 1$, $U_1 = x$, $U_{m+1} = xU_m - U_{m-1}$, so that $U_m(2\cos y) = \sin((m+1)y)/\sin y$; in Exercise 6, π is an involution of the simple objects.

Exercise 1 (Module categories over $\mathrm{Vec}_{\mathbb{Z}_2}$). (i) Using Theorem 11.11, list the indecomposable semisimple module categories over $\mathrm{Vec}_{\mathbb{Z}_2}$, with their simple objects and the matrix n_s.

(ii) The direct sum $\mathrm{Vec}_{\mathbb{Z}_2} \oplus \mathrm{Vect}_{\mathrm{fd}}$ is a module category too. Write its n_s, and say why it is not on the list.

Solution. (i) The subgroups are 1 and $\mathbb{Z}_2$, and $H^2(\mathbb{Z}_2, \mathbb{C}^\times) = 0$ (Example 11.12), so there are two. For $H = 1$, $A = 1$ and $\mathrm{Mod}(A) = \mathrm{Vec}_{\mathbb{Z}_2}$, the regular module, with simple objects δ_1, δ_s; $s \otimes \delta_1 = \delta_s$ and $s \otimes \delta_s = \delta_1$, so $n_s = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. For $H = \mathbb{Z}_2$, $A = \mathbb{C}[\mathbb{Z}_2]$ and $\mathrm{Mod}(A) \simeq \mathrm{Vect}_{\mathrm{fd}}$ with one simple object on which s acts trivially: $n_s = (1)$. (In both, n_1 is the identity.)

(ii) s acts on the three simple objects $\delta_1, \delta_s, \mathbb{C}$ by exchanging the first two and fixing the third, so n_s is block diagonal, $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) \oplus (1)$. The two blocks are module subcategories, so the sum is decomposable; Theorem 11.11 lists the indecomposable ones, of which every finite semisimple module category is a direct sum (EGNO Prop. 7.6.7, cited in Lecture 11).

Exercise 2 (The algebra of a rank-one module). Let ω_p be the cocycle of Sheet 3 on $\mathbb{Z}_4$ and $H = \{0, 2\}$.

(i) Write the module pentagon of Theorem 11.6 at $(g,h,k) = (2,2,2)$ for a normalised ψ, and show that a rank-one module category over the part of $\mathrm{Vec}_{\mathbb{Z}_4}^{\omega_p}$ on H exists exactly when p is even.

(ii) For such a module $\mathcal{M} = \mathrm{Vect}_{\mathrm{fd}}$, with $M = \mathbb{C}$, compute $[M,M]$ by Theorem 11.8 and its product on the basis $e_h \in \mathrm{Hom}(\delta_h \otimes M, M)$, and show that it is the twisted group algebra $\mathbb{C}_\psi[H]$ of Example 11.2.

(iii) For $\mathrm{Mod}(\mathbb{C}[H])$ over Vec_G (trivial ω, $H \leq G$), show that $[M_H, M_H] \cong \mathbb{C}[H]$ as algebras.

Solution. (i) The pentagon of Theorem 11.6 at $(2,2,2)$ reads $\psi(2,2)\,\psi(2+2,2) = \omega_p(2,2,2)\,\psi(2,2)\,\psi(2,2+2)$. As $2+2=0$ and ψ is normalised, $\psi(0,2) = \psi(2,0) = 1$, and dividing by $\psi(2,2) \neq 0$ leaves $1 = \omega_p(2,2,2)$. With the carry $\varepsilon(2,2) = 1$, $\omega_p(2,2,2) = \zeta_4^{2p} = (-1)^p$. The instances with a 0 hold for normalised ψ, and any value of $\psi(2,2)$ then works when p is even. So the module category exists exactly for even p: $\omega_p|_H$ is a coboundary

exactly then (for p odd, $\omega_p|_H$ is the semion's cocycle on $H \cong \mathbb{Z}_2$, which Lecture 11 rules out just after Theorem 11.6).

(ii) By Theorem 11.8, $[M,M] = \bigoplus_c \mathrm{Hom}(c \otimes M, M) \otimes c$. Here $\delta_h \otimes M = M$ for $h \in H$, so $\mathrm{Hom}(\delta_h \otimes M, M) = \mathbb{C}$, spanned by $e_h = \mathrm{id}_M$: $[M,M] = \delta_0 \oplus \delta_2$. The product of e_g and e_h corresponds to $\mathrm{ev} \circ (\mathrm{id} \otimes \mathrm{ev}) \circ \alpha^{\mathcal{M}}$ on $(\delta_g \otimes \delta_h) \otimes M$: the associator $\alpha^{\mathcal{M}}_{g,h,M} = \psi(g,h)$, then e_h, then e_g, so it is the scalar $\psi(g,h)$ times the map $\delta_{g+h} \otimes M \to M$ that corresponds to e_{g+h}: $e_g e_h = \psi(g,h)\, e_{g+h}$. That is the twisted group algebra $\mathbb{C}_\psi[H]$, and its associativity in Vec^{ω_p} (Example 11.2) is the module pentagon of (i). The unit is e_0, from $l^{\mathcal{M}} = \mathrm{id}$.

(iii) Over Vec_G, $A = \mathbb{C}[H] = \bigoplus_{h\in H} \delta_h$ with the group product, and $M_H = A$. Then $\mathrm{Hom}_A(\delta_g \otimes A, A) \cong \mathrm{Hom}(\delta_g, A)$ (free modules), which is $\mathbb{C}$ for $g \in H$ and 0 otherwise: $[M_H, M_H] = \bigoplus_{h\in H} \delta_h$, as an object $\mathbb{C}[H]$. The module map corresponding to the inclusion $\delta_g \to A$ is $e_g(\delta_g \otimes a) = ga$, left multiplication by g. The product of Theorem 11.8 composes evaluations, $e_g \circ (\mathrm{id} \otimes e_h)$, which sends $\delta_g \otimes \delta_h \otimes a$ to $g(ha) = (gh)a$: $e_g e_h = e_{gh}$. So $[M_H, M_H] \cong \mathbb{C}[H]$ as algebras: the internal End of M_H gives back the algebra that $\mathrm{Mod}(\mathbb{C}[H])$ was built from.

Exercise 3 (The four module categories over Vec_{S_3}). You may use that $H^2(H, \mathbb{C}^\times) = 0$ for every subgroup H of S_3.

(i) List the subgroups of S_3 up to conjugacy, and the module categories of Theorem 11.11, with their numbers of simple objects.

(ii) For $H = \mathbb{Z}_3$, write the action of a transposition and of a 3-cycle on the simple objects.

(iii) Over $\mathrm{Rep}(S_3)$ the same pairs give $\mathrm{Rep}(H)$ (Lecture 11, cited). How many simple objects does each have?

Solution. (i) The subgroups are 1; the three subgroups of order 2, $\langle(12)\rangle$, $\langle(13)\rangle$, $\langle(23)\rangle$, all conjugate; $\mathbb{Z}_3 = \langle(123)\rangle$, normal; and S_3. With every H^2 trivial, Theorem 11.11 gives one indecomposable module category per conjugacy class: four, $\mathrm{Mod}(\mathbb{C}[H])$. Its simple objects are the free modules $\delta_g \otimes \mathbb{C}[H]$, one per coset gH (Lecture 11), so the ranks are $|S_3/H| = 6, 3, 2, 1$. For $H = 1$ it is the regular module Vec_{S_3}; for $H = S_3$ it is $\mathrm{Vect}_{\mathrm{fd}}$, the fibre functor forgetting the grading.

(ii) The cosets of $\mathbb{Z}_3$ are $\mathbb{Z}_3$ and $(12)\mathbb{Z}_3$. A 3-cycle maps each coset to itself, a transposition exchanges them: $n_{3\text{-cycle}} = 1$ and $n_{\text{transposition}} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. In general δ_g acts by the permutation $xH \mapsto gxH$ of S_3/H.

(iii) $\mathrm{Rep}(1)$ has one simple object, $\mathrm{Rep}(\mathbb{Z}_2)$ two, $\mathrm{Rep}(\mathbb{Z}_3)$ three and $\mathrm{Rep}(S_3)$ three. The ranks differ from those over Vec_{S_3} (the two lists are matched by the duality of Lecture 12, not rank by rank): $\mathrm{Rep}(1) = \mathrm{Vect}_{\mathrm{fd}}$ is the fibre functor of $\mathrm{Rep}(S_3)$, and $\mathrm{Rep}(S_3)$ is its regular module.

Exercise 4 (NIM-reps; Fibonacci). (i) Show that the NIM-rep of a semisimple module category satisfies $n_1 = 1$ and $n_a n_b = \sum_c N_{ab}^c\, n_c$; you may use that $n_{a^*} = n_a^T$ (duality).

(ii) For Fibonacci, show that a symmetric matrix n of size $r \geq 1$ with non-negative integer entries and $n^2 = 1 + n$ that is indecomposable (no nonempty proper set of simples closed under n) is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ up to relabelling. Which module category has this NIM-rep?

(iii) Let $\mathcal{M}$ be a nonzero indecomposable semisimple module category over Fibonacci, with finitely many simple objects. Show that its NIM-rep is indecomposable. Then take a simple M with $(n_\tau)_{MM} = 0$, and use Theorems 11.8 and 11.10 to show that $\mathcal{M}$ is equivalent to the regular module C.

Solution. (i) $1 \otimes M \cong M$ gives $n_1 = 1$. By semisimplicity $b \otimes M_j \cong \bigoplus_k (n_b)_{kj} M_k$, so $a \otimes (b \otimes M_j)$ contains M_i with multiplicity $\sum_k (n_a)_{ik}(n_b)_{kj} = (n_a n_b)_{ij}$. And $(a \otimes b) \otimes M_j \cong \bigoplus_c N_{ab}^c\, c \otimes M_j$

contains it $\sum_c N_{ab}^c (n_c)_{ij}$ times. The module associator is an isomorphism between the two, so $n_a n_b = \sum_c N_{ab}^c n_c$. With $n_{a^*} = n_a^T$ this is a NIM-rep.

(ii) τ is self-dual, so $n = n_\tau$ is symmetric. The diagonal entry i of $n^2 = 1 + n$ reads $n_{ii}^2 + \sum_{j\neq i} n_{ij}^2 = 1 + n_{ii}$, that is $n_{ii}(n_{ii} - 1) + \sum_{j\neq i} n_{ij}^2 = 1$. If $n_{ii} \geq 2$ the left side is at least 2; so $n_{ii} \in \{0, 1\}$, and then $\sum_{j\neq i} n_{ij}^2 = 1$: exactly one $j \neq i$ with $n_{ij} = 1$, the others 0. So each simple has exactly one neighbour, and indecomposability leaves one pair $\{i, j\}$ (a single simple would need $n_{ii}(n_{ii} - 1) = 1$, impossible). The off-diagonal entry of $n^2 = 1 + n$ reads $n_{ij}(n_{ii} + n_{jj}) = n_{ij}$, so $n_{ii} + n_{jj} = 1$: up to relabelling $n = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, which is N_τ, the NIM-rep of the regular module ($\tau \otimes 1 = \tau$, $\tau \otimes \tau = 1 \oplus \tau$).

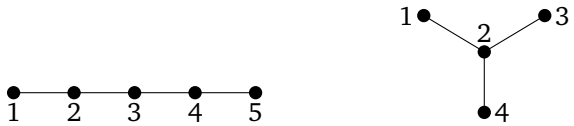
(iii) Suppose a nonempty proper set S of simples is closed under n_τ. As n_τ is symmetric, its complement is closed too, so the direct sums of simples in S, and those of simples not in S, form two nonzero module subcategories (the module associator restricts), and $\mathcal{M}$ is their direct sum: a contradiction. So the NIM-rep is indecomposable, and by (ii) it is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$; let M be the simple with $(n_\tau)_{MM} = 0$. By Theorem 11.8, $[M, M] = \mathrm{Hom}(1 \otimes M, M) \otimes 1 \oplus \mathrm{Hom}(\tau \otimes M, M) \otimes \tau = 1$, as $(n_1)_{MM} = 1$ and $(n_\tau)_{MM} = 0$. An algebra structure on the object 1 is $\mu = \lambda\, l_1$ and $\iota = u$, scalars with $\lambda u = 1$ by the unit law, and the scalar $\lambda : 1 \to 1$ is an isomorphism of algebras onto $(1, l_1, \mathrm{id}_1)$: $\lambda \cdot \lambda = \lambda^2$ and $\lambda u = 1$. By Theorem 11.10, $\mathcal{M} \simeq \mathrm{Mod}_C(1)$, which is all of C with its own action (Example 11.2): the regular module.

Exercise 5 (SU(2)$_4$: A_5 and D_4). For SU(2)$_4$, a NIM-rep is determined by $n = n_{1/2}$ through $n_0 = 1$ and $n_{j+1/2} = n\, n_j - n_{j-1/2}$ (Sheet 5, Exercise 2).

(i) The Dynkin diagram A_5 is the path on the vertices $1, 2, 3, 4, 5$, and D_4 is the graph with a centre 2 joined to three ends $1, 3, 4$. Draw them, and let n be the adjacency matrix of either. Show that the n_j, $j = 0, \ldots, 2$, are non-negative integer matrices and that $n_{5/2} = 0$, and deduce that the n_j form a NIM-rep.

(ii) Show that the eigenvalues of n are $2\cos(m\pi/6)$ for $m = 1, \ldots, 5$ (A_5) and $m = 1, 3, 3, 5$ (D_4), and explain why $n_{5/2} = 0$ follows from this.

Solution. (i) A_5 is the path $1 - 2 - 3 - 4 - 5$; D_4 has a centre joined to three ends:



The recursion gives $n_1 = n^2 - 1$, $n_{3/2} = n^3 - 2n$, $n_2 = n^4 - 3n^2 + 1$ and $n_{5/2} = n^5 - 4n^3 + 3n$. For A_5:

$$n_1 = \left(\begin{smallmatrix} 0&0&1&0&0 \\ 0&1&0&1&0 \\ 1&0&1&0&1 \\ 0&1&0&1&0 \\ 0&0&1&0&0 \end{smallmatrix}\right), \quad n_{3/2} = \left(\begin{smallmatrix} 0&0&0&1&0 \\ 0&0&1&0&1 \\ 0&1&0&1&0 \\ 1&0&1&0&0 \\ 0&1&0&0&0 \end{smallmatrix}\right), \quad n_2 = \left(\begin{smallmatrix} 0&0&0&0&1 \\ 0&0&0&1&0 \\ 0&0&1&0&0 \\ 0&1&0&0&0 \\ 1&0&0&0&0 \end{smallmatrix}\right),$$

and $n_{5/2} = n\, n_2 - n_{3/2} = 0$, since $n\, n_2$ is $n_{3/2}$ (the path reversed). For D_4, with the centre second:

$$n_1 = \left(\begin{smallmatrix} 0&0&1&1 \\ 0&2&0&0 \\ 1&0&0&1 \\ 1&0&1&0 \end{smallmatrix}\right), \quad n_{3/2} = n, \quad n_2 = 1,$$

and $n_{5/2} = n\, n_2 - n_{3/2} = n - n = 0$. All entries are non-negative integers.

These are a NIM-rep. By Sheet 5, Exercise 2(ii), $[j] = U_{2j}([\tfrac{1}{2}])$, so $x \mapsto [\tfrac{1}{2}]$ maps $\mathbb{Z}[x]$ onto $K(\mathrm{SU}(2)_4)$; it kills U_5, since $U_5([\tfrac{1}{2}]) = [\tfrac{1}{2}][2] - [\tfrac{3}{2}] = 0$ by the truncated rule. Both $\mathbb{Z}[x]/(U_5)$ (U_5 is monic of degree 5) and $K(\mathrm{SU}(2)_4)$ are free of rank 5, so $K(\mathrm{SU}(2)_4) \cong \mathbb{Z}[x]/(U_5)$. As $U_5(n) = n_{5/2} = 0$, $x \mapsto n$ is a ring map from $K(\mathrm{SU}(2)_4)$, sending $[j]$ to $U_{2j}(n) = n_j$: so $n_0 = 1$ and $n_a n_b = \sum_c N_{ab}^c n_c$. The n_j are symmetric and every j is self-dual, so they form a NIM-rep. (The script also checks every product.)

(ii) The eigenvalues of the path A_5 are $2\cos(m\pi/6)$, $m = 1, \dots, 5$ (the eigenvectors are $(\sin(mi\pi/6))_i$). For D_4: the vector $(1, 0, -1, 0)$ and $(1, 0, 0, -1)$ on the ends have eigenvalue $0 = 2\cos(3\pi/6)$, twice; on vectors constant on the ends, n acts as $\left(\begin{smallmatrix} 0 & 3 \\ 1 & 0 \end{smallmatrix}\right)$ (centre, end), with eigenvalues $\pm\sqrt{3} = 2\cos(\pi/6), 2\cos(5\pi/6)$. Since n is symmetric, $n_{5/2} = U_5(n)$ vanishes exactly when U_5 vanishes on every eigenvalue: $U_5(2\cos y) = \sin 6y/\sin y$ is 0 at $y = m\pi/6$, $m = 1, \dots, 5$. The largest eigenvalue is $\sqrt{3} = d_{1/2}$ (Sheet 5). The module categories with these NIM-reps exist; Ostrik (math/0111139) classifies those over $\mathrm{SU}(2)_k$ by the ADE diagrams (cited, a preview).

Exercise 6 ⋆. *Ising has only the regular module.* A NIM-rep of the Ising ring is a pair of symmetric $r \times r$ ($r \geq 1$) non-negative integer matrices n_σ, n_ψ with $n_\psi^2 = 1$, $n_\sigma n_\psi = n_\sigma$ and $n_\sigma^2 = 1 + n_\psi$.

(i) Show that n_ψ is the permutation matrix of an involution π of the simple objects, and that each row of n_σ is unchanged by π.

(ii) Show that if π moves i, then row i of n_σ has a single 1, at a point j that π fixes, and row j is $e_i + e_{\pi(i)}$.

(iii) Show that π cannot fix every simple object, and conclude that every indecomposable NIM-rep is the regular one, with three simple objects.

(iv) Conclude, as in Exercise 4(iii), that every nonzero indecomposable semisimple module category over Ising, with finitely many simple objects, is equivalent to the regular one.

Solution. (i) n_ψ is its own inverse, and a non-negative integer matrix with a non-negative integer inverse is a permutation matrix (each row of $n_\psi n_\psi = 1$ forces exactly one non-zero entry, equal to 1, in each row and column). It is symmetric, so the permutation π is an involution. $(n_\sigma n_\psi)_{ij} = (n_\sigma)_{i\pi(j)}$, so $n_\sigma n_\psi = n_\sigma$ says that each row of n_σ is constant on the orbits of π. Transposing, $n_\psi n_\sigma = n_\sigma$: rows i and $\pi(i)$ of n_σ are equal.

(ii) The diagonal of $n_\sigma^2 = 1 + n_\psi$ reads $\sum_k (n_\sigma)_{ik}^2 = 1 + [\pi(i) = i]$. If $\pi(i) \neq i$ the sum is 1: row i has a single 1, at some j. The row is constant on π-orbits, so $\pi(j) = j$ (otherwise there would be a second 1 at $\pi(j)$). By symmetry $(n_\sigma)_{ji} = 1$, and rows i and $\pi(i)$ are equal, so $(n_\sigma)_{j\pi(i)} = 1$ too. As $\pi(j) = j$, row j has square-sum 2, which these two 1's exhaust: row j is $e_i + e_{\pi(i)}$.

(iii) Suppose π fixes every simple of an indecomposable NIM-rep. Then $n_\psi = 1$ and $n_\sigma^2 = 2$: every row of n_σ has square-sum 2, so two entries equal to 1, say at $j \neq k$, and every off-diagonal entry of n_σ^2 is 0. If $j = i$ (a loop), then $(n_\sigma^2)_{ik} \geq (n_\sigma)_{ii}(n_\sigma)_{ik} = 1$; if $j, k \neq i$, then $(n_\sigma^2)_{jk} \geq (n_\sigma)_{ji}(n_\sigma)_{ik} = 1$. Either way an off-diagonal entry is non-zero: a contradiction. So π moves some i, and by (ii) the set $\{i, \pi(i), j\}$ is closed under n_σ and n_ψ: rows $i, \pi(i)$ point only to j, row j only to $i, \pi(i)$, and π swaps $i, \pi(i)$ and fixes j. By indecomposability this is everything: $n_\sigma = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right)$ and n_ψ swapping the ends, with j in the middle; that is N_σ and N_ψ on $(1, \sigma, \psi)$, the regular NIM-rep. (The script confirms there is no other up to five simple objects.)

(iv) Every simple of Ising is self-dual, so n_σ and n_ψ are symmetric, and the argument of Exercise 4(iii) shows that the NIM-rep of an indecomposable $\mathcal{M}$ is indecomposable: by (iii), the regular one. At a point i moved by π, $(n_\psi)_{ii} = 0$, and row i of n_σ has its single 1 at $j \neq i$, so $(n_\sigma)_{ii} = 0$. By Theorem 11.8, $[M_i, M_i] = 1$, an algebra isomorphic to 1 as in Exercise 4(iii), and Theorem 11.10 gives $\mathcal{M} \simeq \mathrm{Mod}_C(1) = C$.